Calculus II Lesson 31: Polar Coordinates

Rectangular and Polar Systems

$x = r\cos\theta; \quad y = r\sin\theta$

$x^2 + y^2 = r^2; \quad r = \pm\sqrt{x^2 + y^2}$

$\tan\theta = \dfrac{y}{x}; \quad \theta = \tan^{-1}\left(\dfrac{y}{x}\right)$

How to use calculator to figure out θ :

Quadrant II $x < 0;\ y > 0$ $\theta = \tan^{-1}(y/x) + \pi$	Quadrant I $x > 0;\ y > 0$ $\theta = \tan^{-1}(y/x)$
$x < 0;\ y < 0$ $\theta = \tan^{-1}(y/x) + \pi$ Quadrant III	$x > 0;\ y < 0$ $\theta = \tan^{-1}(y/x) + 2\pi$ Quadrant IV

Also:

If (x,y) is on positive y – axis ($x = 0$ and $y > 0$): $\theta = \pi/2$.

If (x,y) is on negative y – axis ($x = 0$ and $y < 0$): $\theta = 3\pi/2$.

If (x,y) is on positive x – axis ($x > 0$ and $y = 0$): $\theta = 0$.

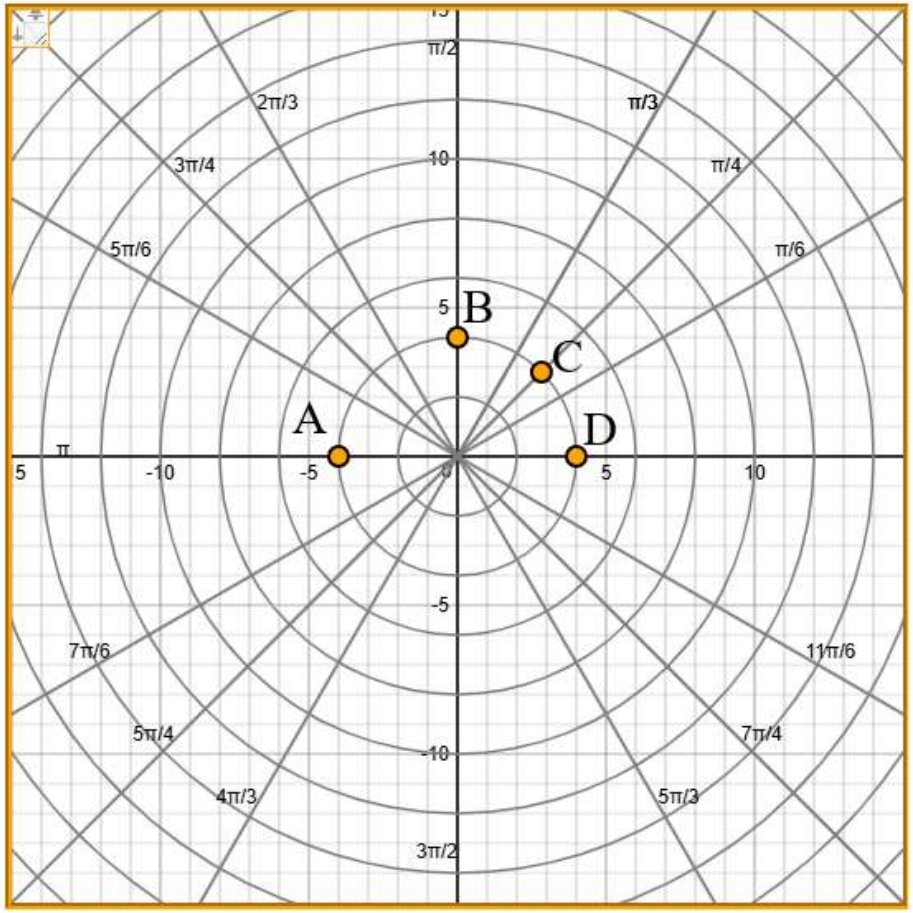
If (x,y) is on negative x – axis ($x < 0$ and $y = 0$): $\theta = \pi$.

1) a) Plot polar point $\left(4, \frac{\pi}{2}\right)$ on the above grid.

(multiple choice). Choose the correct answer (A, B, C, or D).

b) Write Corresponding Rectangular Coordinates

$x =$ ___?___ and $y =$ ___?___



Answers

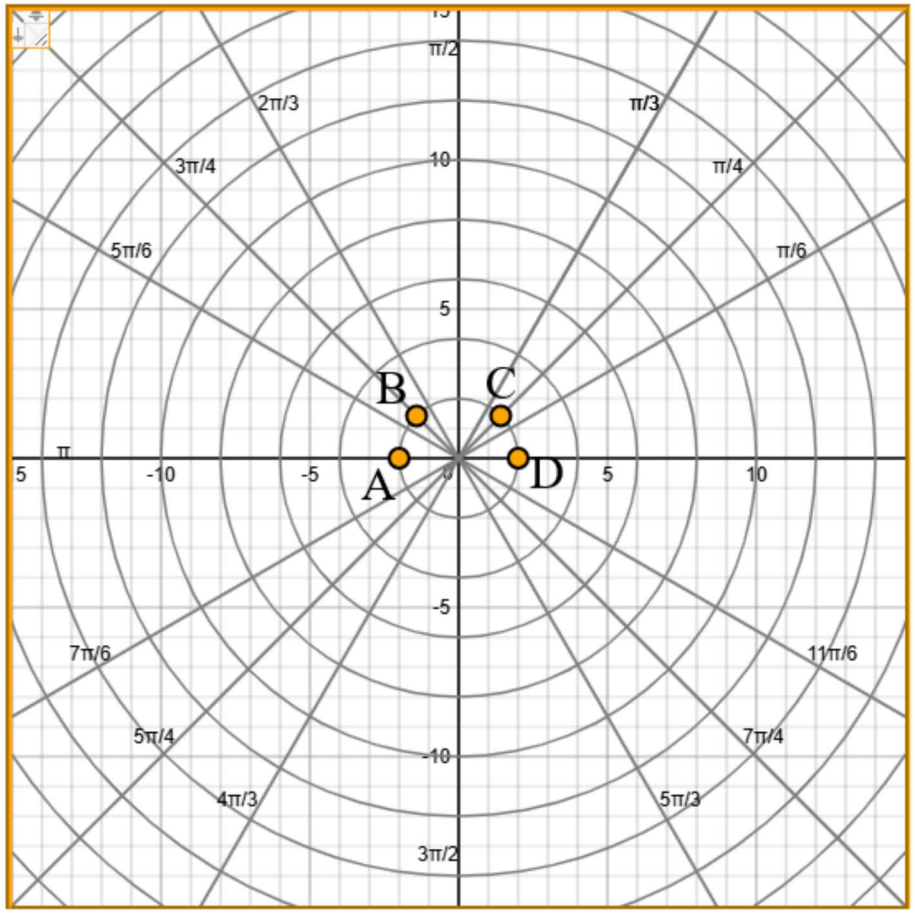
a) b)

2) a) Plot polar point $\left(-2,-\frac{3\pi}{4}\right)$ on the above grid.

(multiple choice). Choose the correct answer (A, B, C, or D).

b) Write Corresponding Rectangular Coordinates

$x =$ ___?___ and $y =$ ___?___

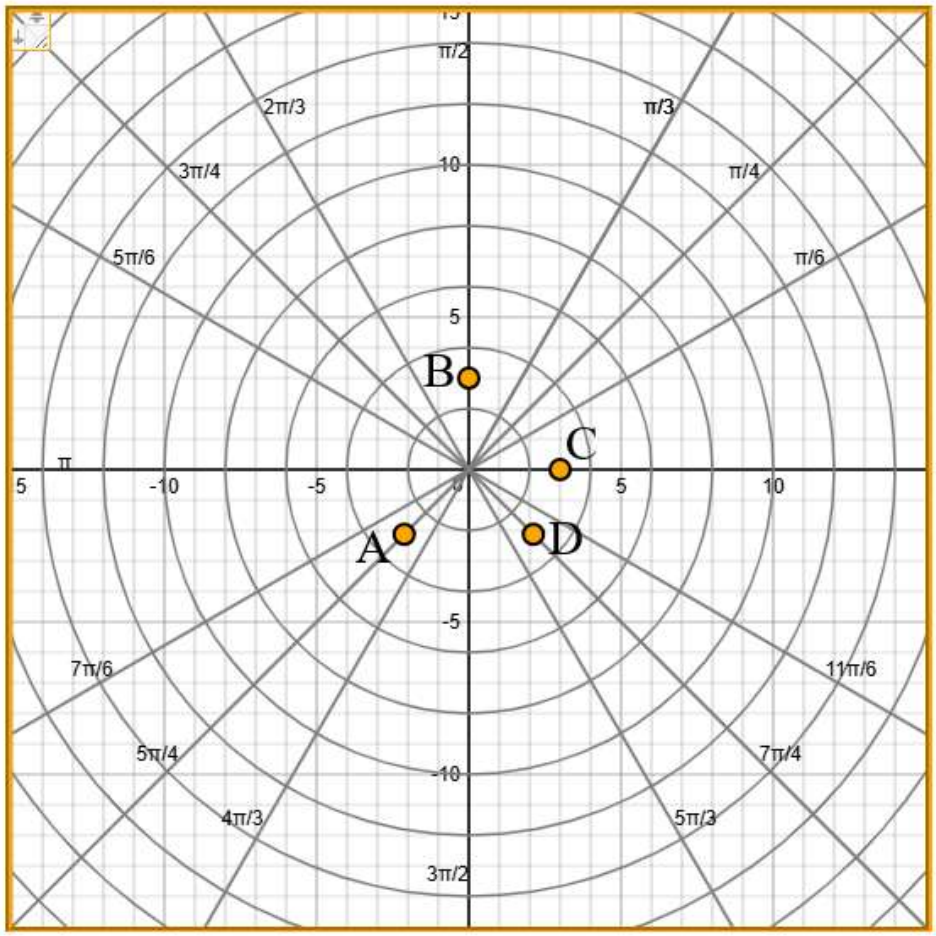


Answers

a) b)

3) a) Plot polar point $\left(3, -\frac{3\pi}{4}\right)$ on the above grid.

(multiple choice). Choose the correct answer (A, B, C, or D)



b) Write Corresponding Rectangular Coordinates

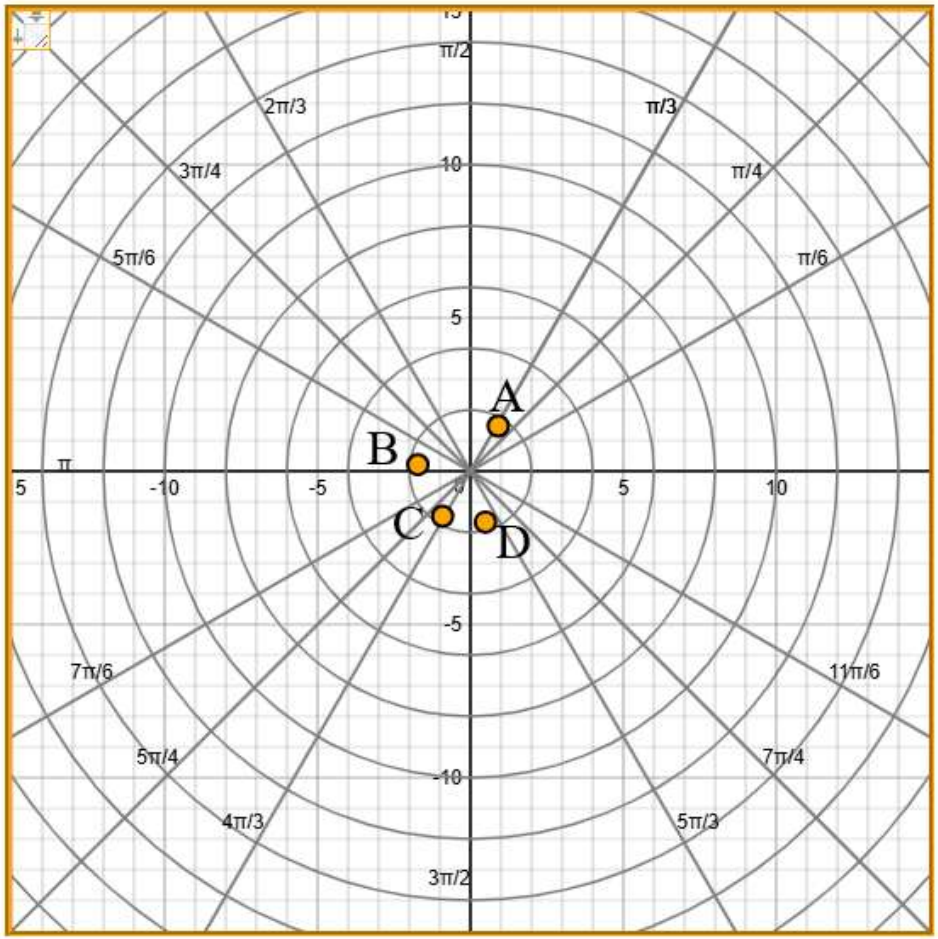
$x =$ ___?___ and $y =$ ___?___

Answers

a) b)

4) a) Plot polar point $\left(\sqrt{3}, 4.16\right)$ on the above grid.

(multiple choice). Choose the correct answer (A, B, C, or D)



b) Write Corresponding Rectangular Coordinates

$x =$ ___?___ and $y =$ ___?___

Answers

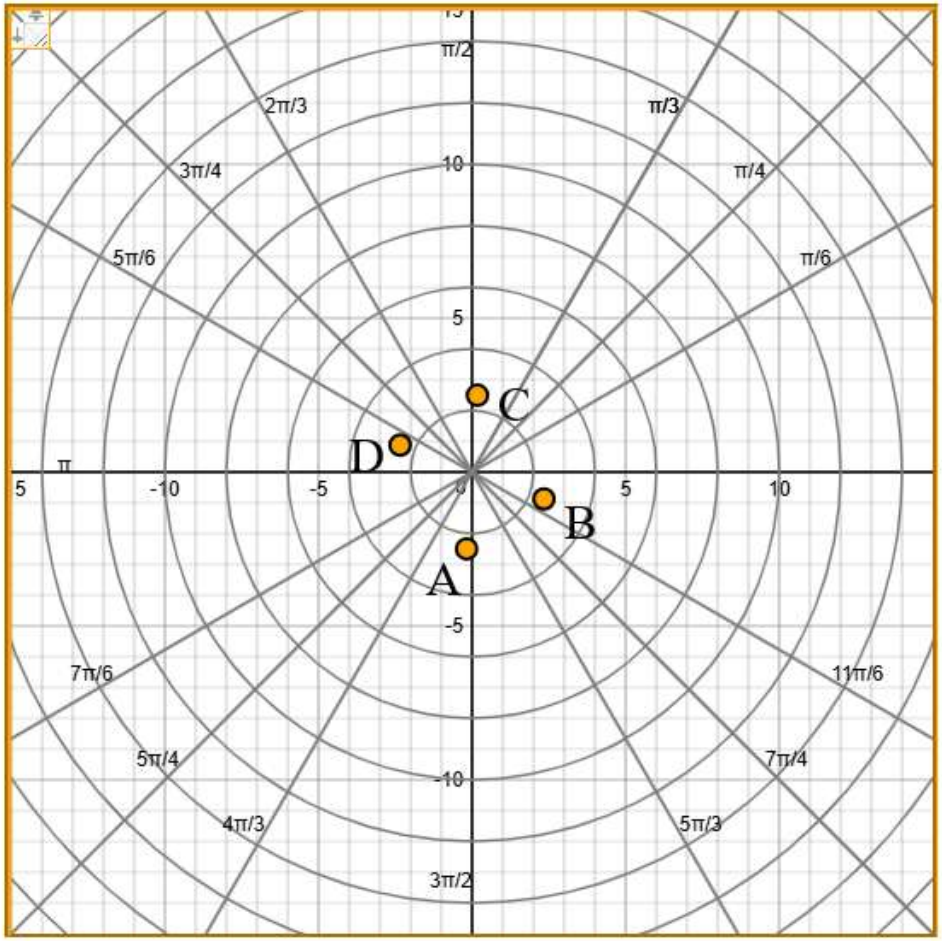
a) b)

5) a) Plot polar point $(-2.5, 1.5)$ on the above grid.

(multiple choice). Choose the correct answer (A, B, C, or D)

b) Write Corresponding Rectangular Coordinates

$x =$ ___?___ and $y =$ ___?___



Answers

a) b)

6) Rectangular Coordinates $(x, y) = (5,3)$.

Write two sets of corresponding polar coordinates for $(5,3)$.

a) ? _____

b) ? _____

Note: $(5,3)$ is a point in the Quadrant I,

So $\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{5}\right)$

Answers

a)

b)

7) Rectangular Coordinates $(x, y) = (-2, 3)$.

Write two sets of corresponding polar coordinates for $(-2, 3)$.

a) ? _____

b) ? _____

Note: $(-2, 3)$ is a point in the Quadrant II,

So $\theta = \tan^{-1}\left(\frac{y}{x}\right) + \pi = \tan^{-1}\left(\frac{3}{-2}\right) + \pi$

Answers

a)

b)

8) Rectangular Coordinates $(x, y) = \left(-2, \sqrt{5}\right)$.

Write two sets of corresponding polar coordinates for $\left(-2, \sqrt{5}\right)$.

a) ? _____

b) ? _____

Note: $\left(-2, \sqrt{5}\right)$ is a point in the Quadrant II,

So $\theta = \tan^{-1}\left(\frac{y}{x}\right) + \pi = \tan^{-1}\left(\frac{-2}{\sqrt{5}}\right) + \pi$

Answers

a)

b)